WEEK 07 – DIMENSIONALITY REDUCTION

Moneyball: IPL Player Valuation | Team 05
Course: 24ADI204 – Data Science and Visualization

Team Members
Divyesh D S – 25BAD022 Kavi Prakash M – 25BAD043
Mithuin P S – 25BAD057 Madhan P M – 25BAD503
Renganathan G M – 25BAD087

1. Objective

The objective of Week 7 is to reduce the dimensionality of the IPL player dataset while retaining as much information as possible. The workflow includes numerical feature selection, correlation analysis, multicollinearity reduction, standard scaling and Principal Component Analysis (PCA).

2. Dataset Overview

The analysis used the Week 3 cleaned IPL dataset containing **3,311 rows** and **27 columns**. The source dataset contains 17 numerical columns before excluding the season identifier. The report uses only this cleaned dataset; Week 6 output CSV files are not required.

3. Numerical Feature Preparation

Numerical columns were selected and the 'season' identifier was excluded. Missing numerical values were imputed using the median. Constant numerical features were removed. After these steps, **16** varying numerical features remained for correlation analysis. The original source DataFrame was not overwritten.

4. Correlation Analysis and Heatmap

Pearson correlation was calculated for the selected numerical features. The heatmap below visualizes the strength and direction of the relationships. A threshold of **0.90** was used to identify highly correlated feature pairs for redundancy reduction.

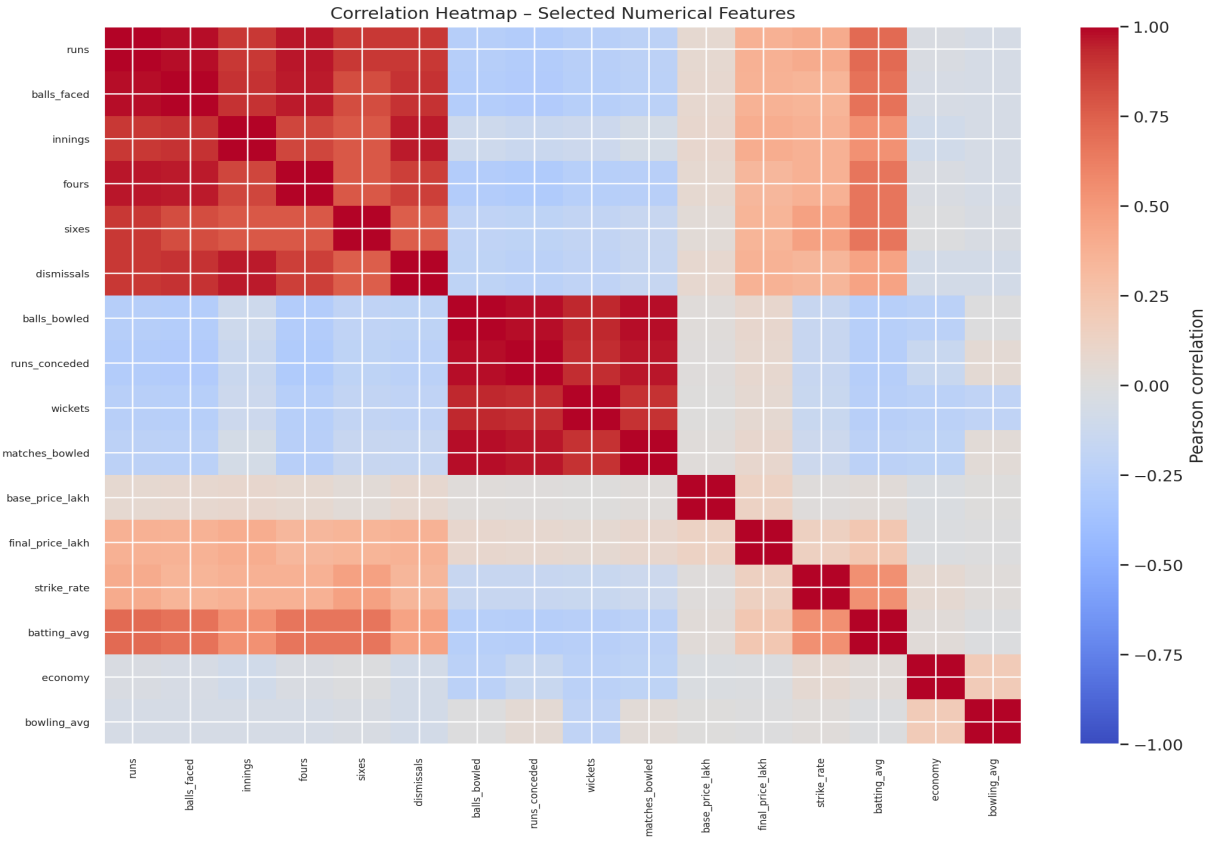


Figure 1. Correlation heatmap of selected numerical features.

5. Multicollinearity Reduction

Using the upper triangle of the absolute correlation matrix and a threshold greater than 0.90, the following features were removed: **balls_faced, innings, fours, dismissals, runs_conceded, wickets, matches_bowled**. The number of features decreased from **16** to **9**. This is a simple correlation-based filter and does not replace domain-based feature review.

6. Standard Scaling

The remaining numerical features were standardized using StandardScaler. Standardization puts features on comparable scales by centering them around zero and scaling them to unit variance. This is important because PCA is sensitive to differences in feature scales.

7. Principal Component Analysis

PCA was fitted to the standardized data. The full PCA model produced **9 principal components**. The smallest number of components required to reach the 95% cumulative variance target was **7**. The cumulative variance retained by these components is **95.18%**.

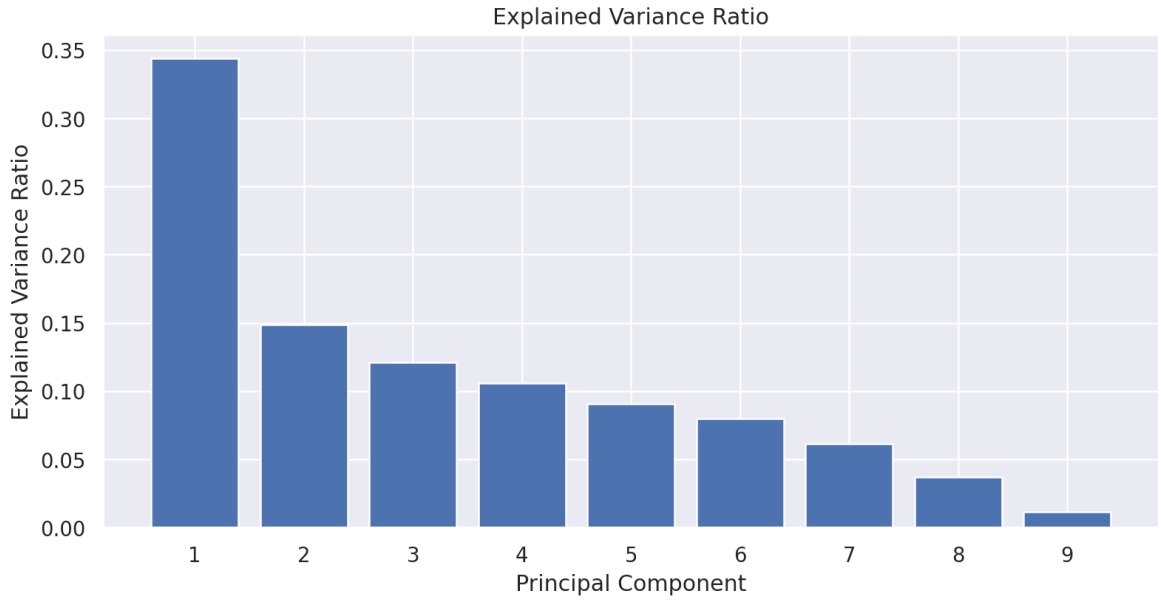


Figure 2. Explained variance ratio for each principal component.

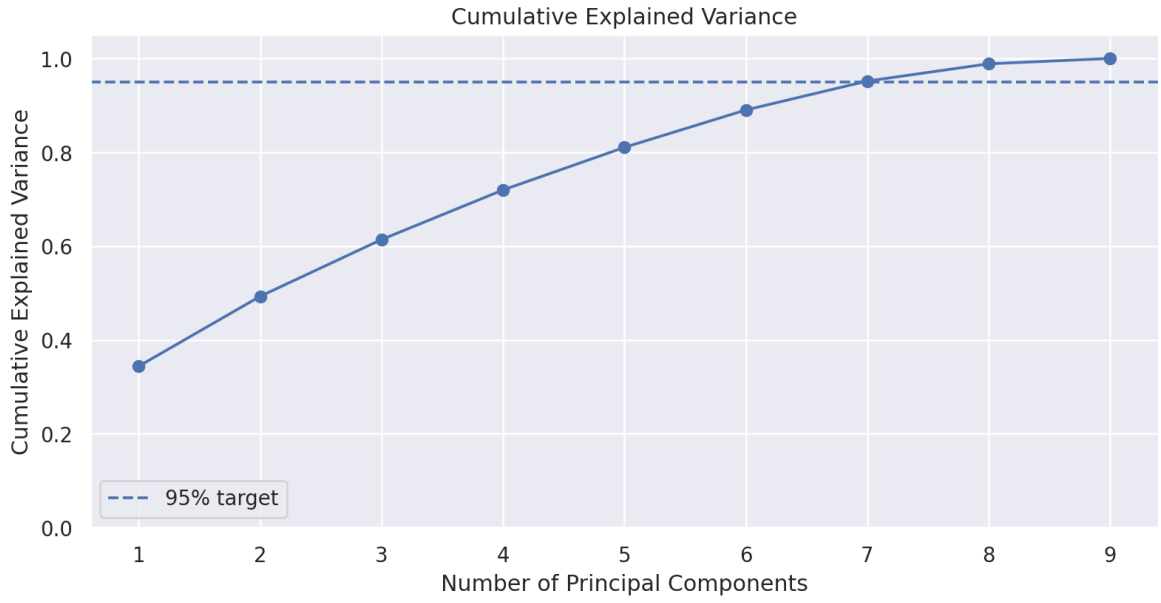


Figure 3. Cumulative explained variance and 95% target.

8. PCA Visualizations

The first two principal components were plotted to provide a two-dimensional view of the observations. The 3D plot uses the first three principal components when at least three numerical features are available. These plots show the data in a compressed coordinate space; they do not by themselves indicate player quality or auction value.

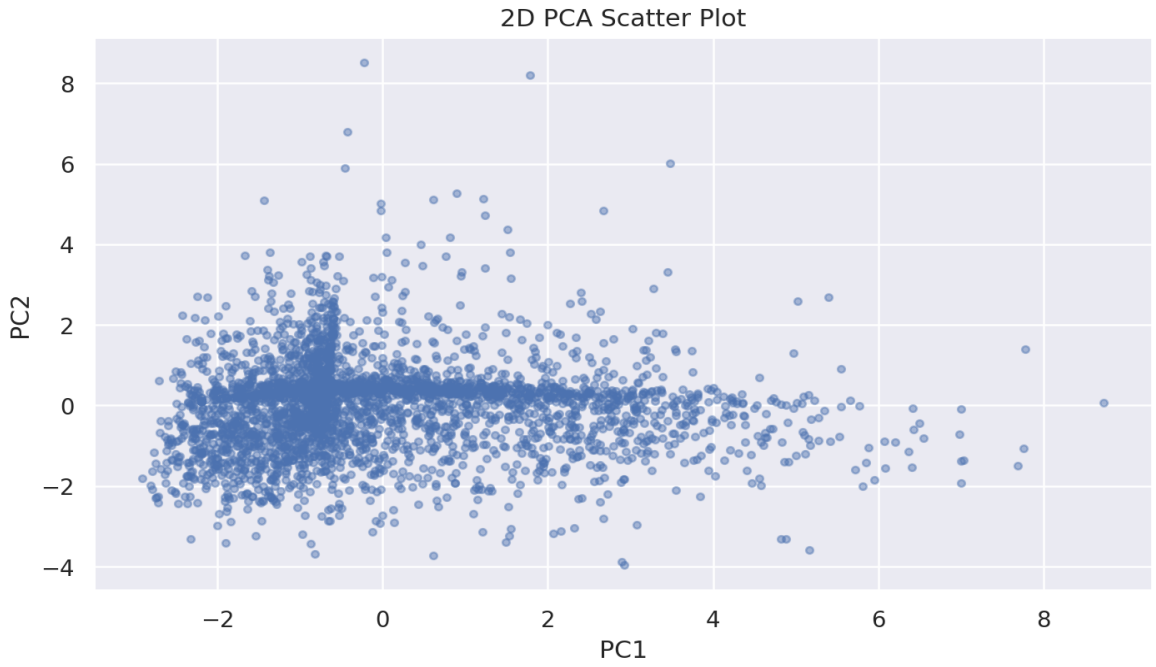


Figure 4. 2D PCA scatter plot (PC1 against PC2).

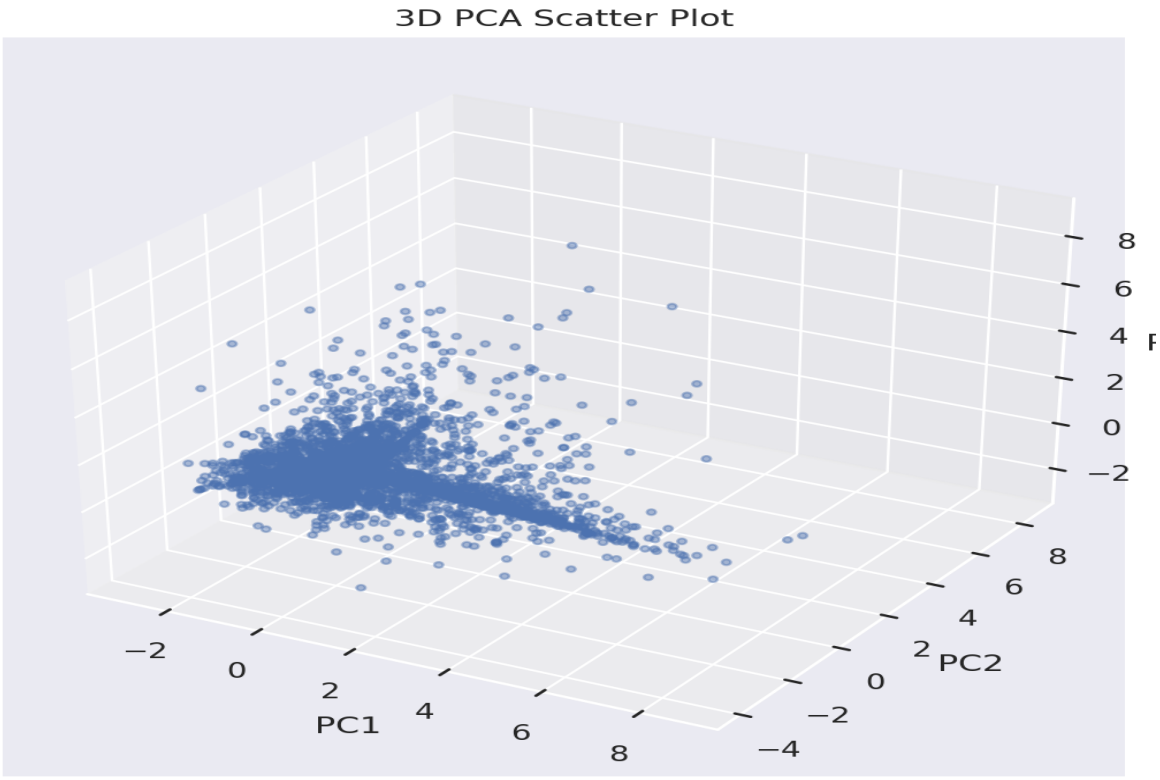


Figure 5. 3D PCA scatter plot (PC1, PC2 and PC3).

9. Results Summary

Measure	Result
Original dataset	3311 rows × 27 columns

Measure	Result
Numerical features after constant removal	16
Features after correlation filtering	9
Principal components retained	7
Variance retained	95.18%
Correlation threshold	0.9

10. Conclusion

Week 7 dimensionality reduction was performed using the cleaned IPL dataset. Numerical features were prepared through median imputation and constant-feature removal. Correlation analysis was used to identify and remove highly correlated features, followed by standard scaling and PCA. The selected 7 components retain 95.18% of the variance. The generated plots provide visual evidence of feature relationships and the variance represented by the principal components.

11. Output Files

The notebook saves two output files: **IPL_PCA_Reduced_Week07.csv** (the reduced PCA representation) and **IPL_Correlation_Matrix_Week07.csv** (the correlation matrix before high-correlation feature removal).

Note: The numerical values and graphs in this report were calculated from the uploaded Week 3 cleaned CSV. Results may vary if the input dataset is changed.